A Taxonomy of Treemap Visualization Techniques

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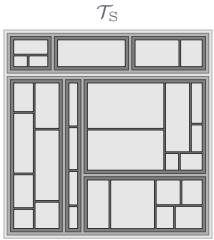
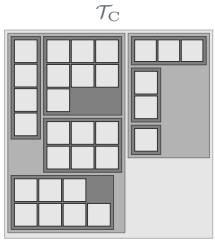
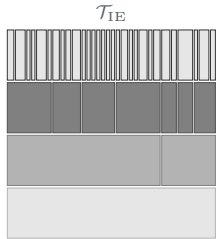
Hasso Plattner Institute, Digital Engineering Faculty, University of Potsdam



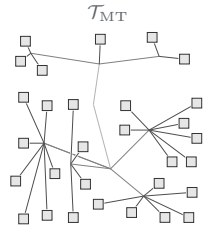
Space-filling Treemap



Containment Treemap

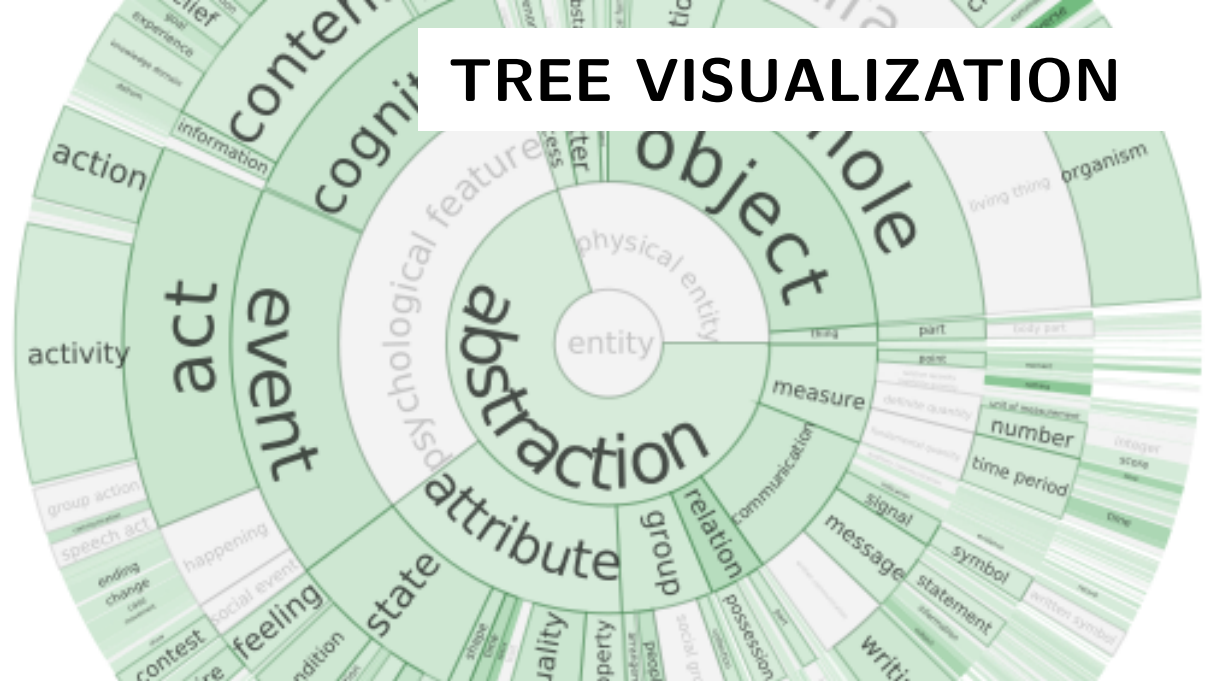


Implicit Edge Representation Tree



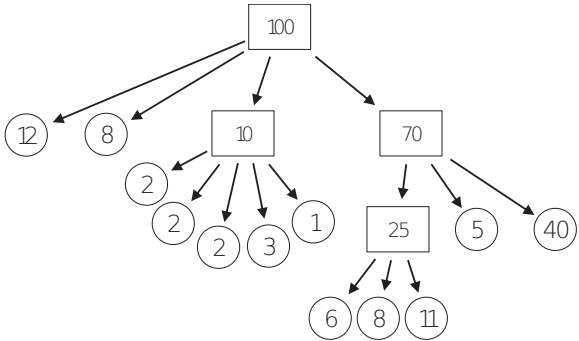
Mapped Tree

IVAPP 2020
Saturday 29th February, 2020


TREE VISUALIZATION
entity
physical entity
psychological feature
abstraction
object
event
act
action
activity
attribute
state
feeling
relation
group
communication
measure
number
time period
signal
message
symbol
statement
part
organism
living thing
information
belief
experience
contest
ending
change
happening
social event
group action
speech act
possession

Tree Visualization

- Visualization of tree-structured data
- Besides relationships of nodes, often numerical or categorical measures are of interest, too.
- Diverse use in information visualization



Example of a tree-structured dataset with associated weights for leaf nodes and accumulated weights for inner nodes.

Overview of Tree Visualization Techniques

Examples of tree visualization techniques (excerpt from treevis.net [Schulz, 2011])

Problem Statement

- When is a visualization technique a treemap?
- What are characteristics of treemaps?

- Quick answer: it is based “on the property of containment” [Johnson, 1993]

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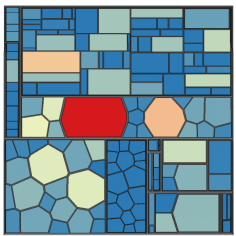
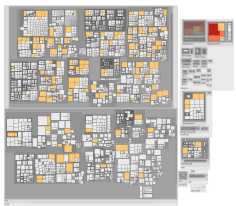
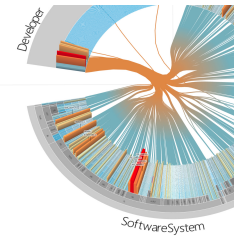
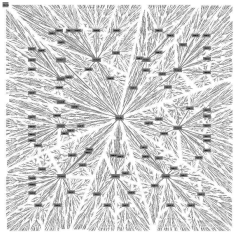


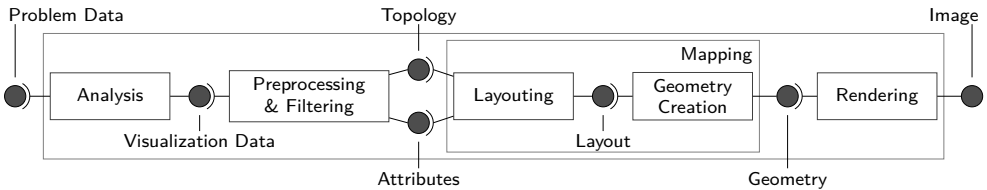






Tree Visualization Pipeline Model

- Abstract model of a visualization pipeline for tree-structured data with the intend to generate tree visualizations.
- Important concepts are the explicit use of the topology and additional attributes for layouting and geometry processing.



Pipeline model for tree visualization techniques (including treemaps), adopted from [dos Santos and Brodlie, 2004].

TREEMAP

Treemaps [Johnson and Shneiderman, 1991] are used to visualize tree-structured data,

e.g., in demographics [Jern et al., 2009], business intelligence [Roberts and Laramee, 2018], health [Chazard et al., 2006], and software development [Merino et al., 2018].

1. non-spatial data is spatialized – it is given a gestalt that preserves the data's structure, e.g., rectangular treemaps use nested rectangles to depict nodes,
2. and data (attributes) are mapped to visual variables, i.e., properties of the rectangles (size and color),
3. using one or multiple metaphors for value encoding [Limberger et al., 2019]

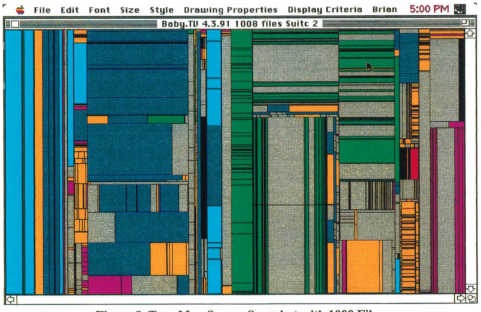


Figure 8. Tree-Map Screen Snapshot with 1000 Files

Characteristics of a Treemap

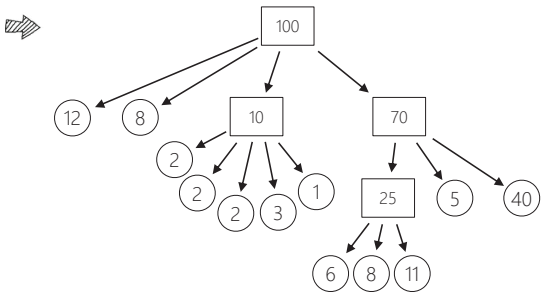
A visualization technique is called a treemap, especially if:

1. A treemap is a visualization technique suitable to depict tree-structured data.
2. There is a representation of the parent-child relationships without use of explicit edge links, i.e., the containment metaphor [Schulz et al., 2011].
3. The depiction of leaf nodes implies an area or volume in the geometry space [Park et al., 2018].
4. The representation of sibling nodes should be guaranteed to be non-overlapping [Johnson and Shneiderman, 1991].

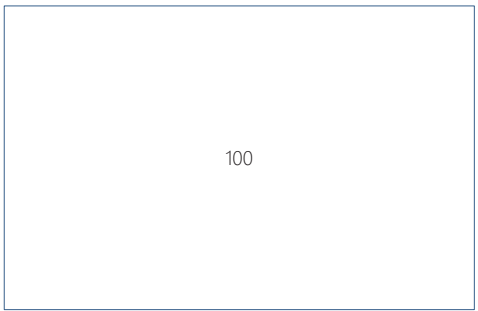
Basic Treemap Layout Example

Leaf nodes are colored and have numerical weights, which can represent any associated attribute, e.g., file size. The weight of a parent node is defined by the sum of the weights of its child nodes.

Simple Graph



Rectangular Treemap



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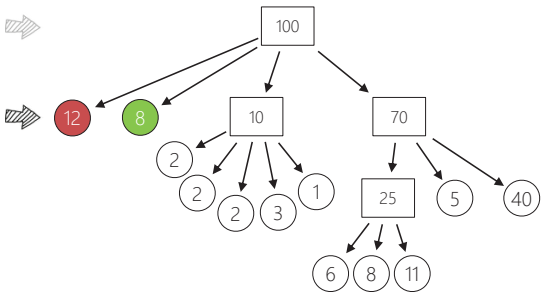
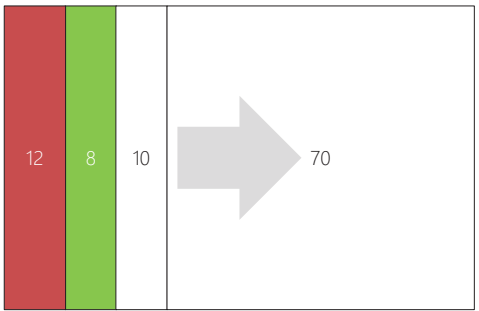


Rectangular Treemap



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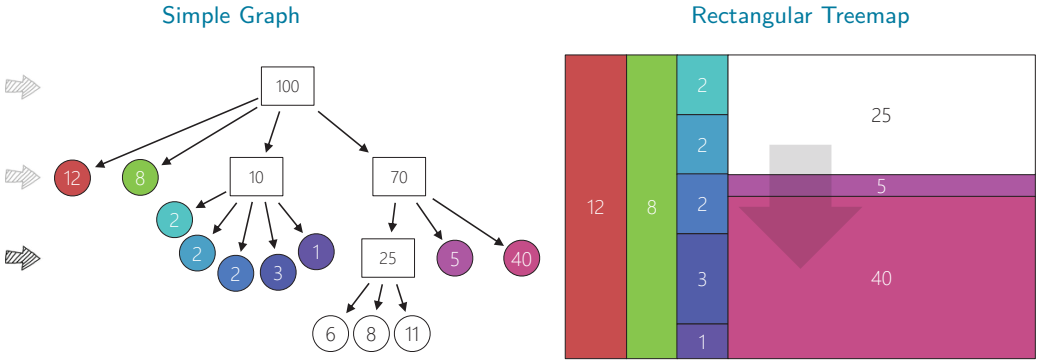

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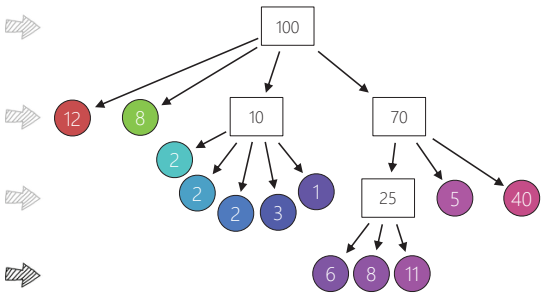
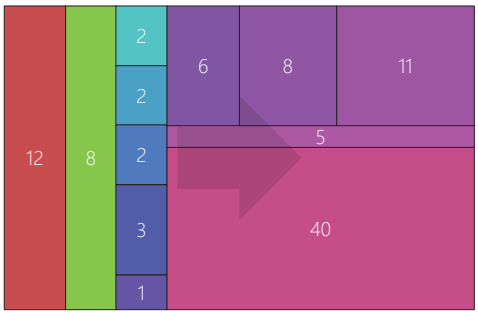


Rectangular Treemap



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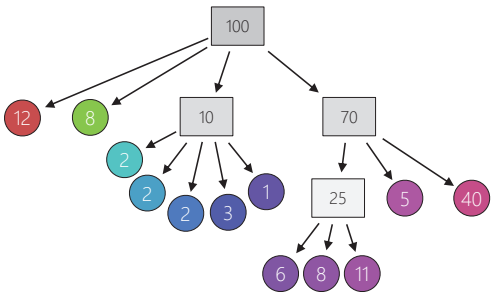
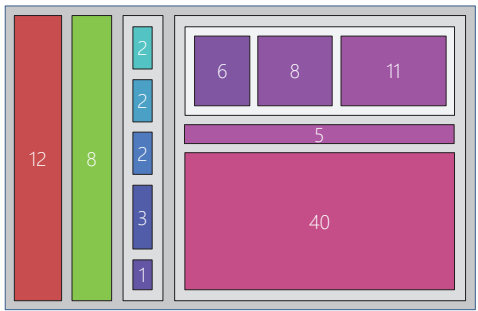
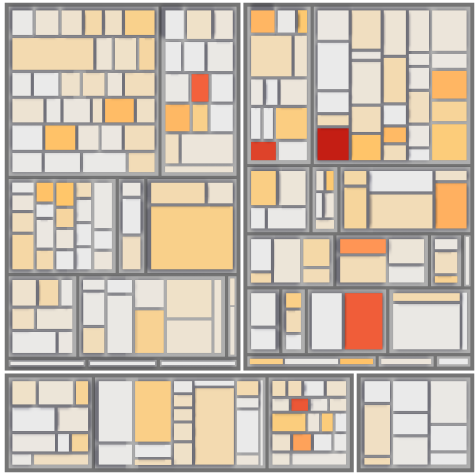


Rectangular Treemap



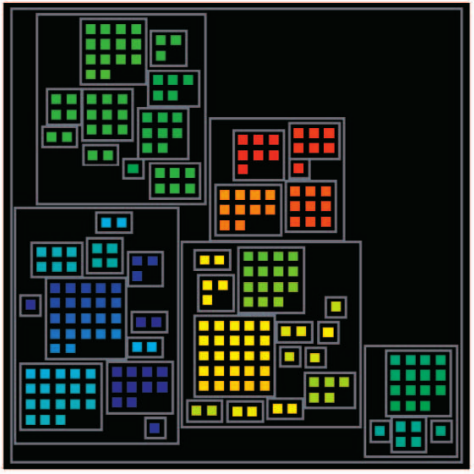
Splitting Treemap Layouts

- Full subdivision of a parent's layout [Schulz et al., 2011]
- Abstract layout algorithm for splitting layouts
  1. Assign layout for root level
  2. Assign root as current node
  3. Layout all children of current node within layout of current node
  4. Repeat 2–3 for all children nodes as current node
- $\Rightarrow$ a top-down approach, iterating from root to leaf nodes

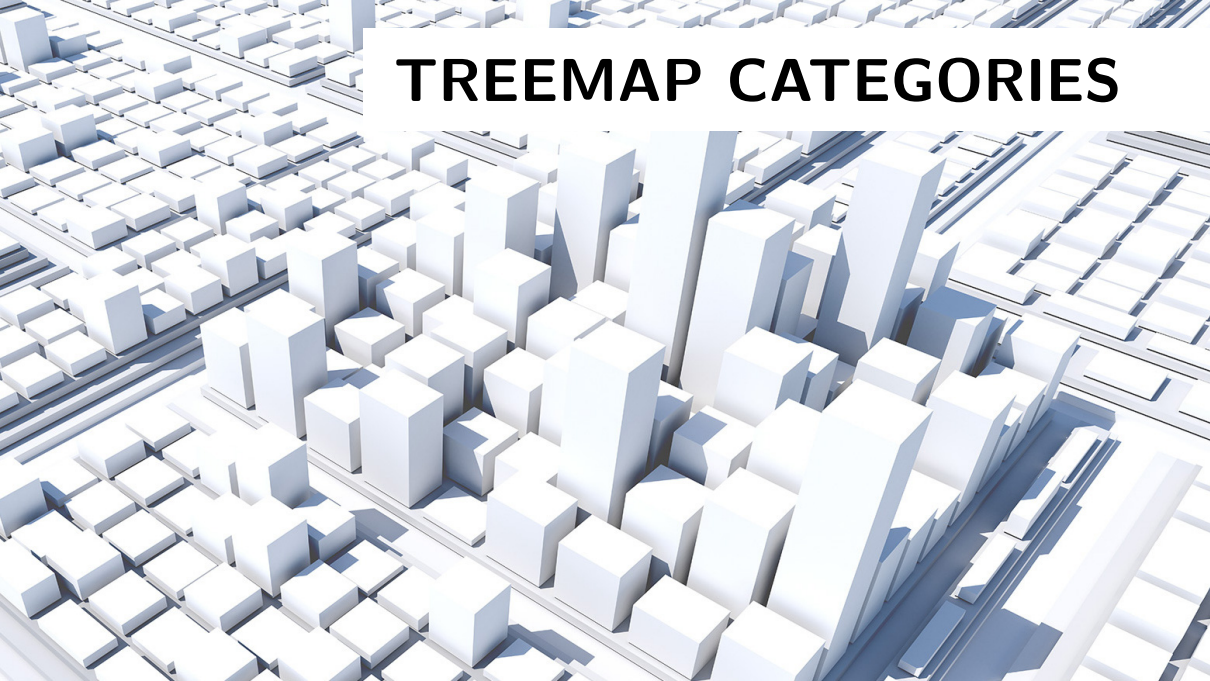


Packing Treemap Layouts

- Derivation of parent layout from child layouts [Schulz et al., 2011]
- Abstract layout algorithm for packing layouts
  1. Assign root as current node
  2. Repeat 2–4 for all children nodes as current node
  3. If current node is leaf node, assign layout
  4. If current node is inner node, determine layout by placement of all child layouts and derivation of bounding box as own layout
- $\Rightarrow$ a bottom-up approach, iterating from leaf nodes to root node

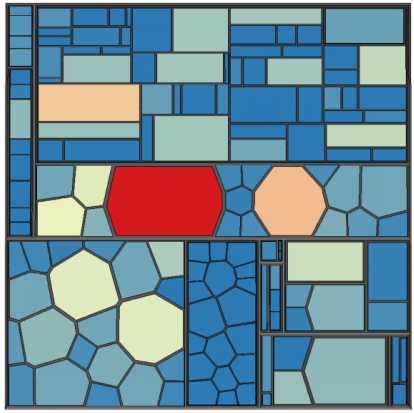


TREEMAP CATEGORIES

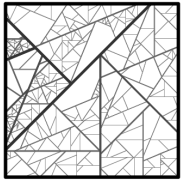


Splitting Treemap – $\mathcal{T}_S$

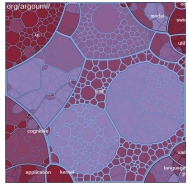
$\mathcal{T}_S$



Hybrid Layout Treemap



Treemap with Bound Aspect Ratio



Voronoi Treemap

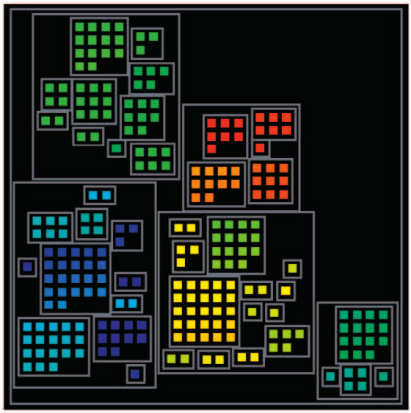
GosperMap

Jigsaw Treemap

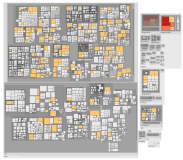
Examples of treemaps from the category *Space-Filling Treemap*. Image courtesy from left to right and top to bottom: [Hahn and Döllner, 2017], [de Berg et al., 2011], [Balzer et al., 2005], [Auber et al., 2013], and [Wattenberg, 2005].

Packing Treemap – $\mathcal{T}_C$

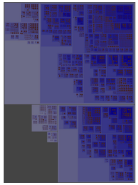
$\mathcal{T}_C$



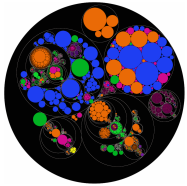
Data-Jewelry Box



EvoCells Treemap



CodeCity



Pebbles Treemap

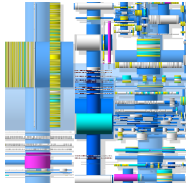
Rectangular Hierarchical Cartogram

Examples of treemaps from the category *Containment Treemap*. Image courtesy from left to right and top to bottom: [Itoh et al., 2004], [Scheibel et al., 2018], [Wettel and Lanza, 2007], [Wetzel, 2003], and [Slingsby et al., 2010].

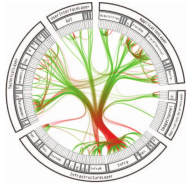
Implicit Edge Representation Tree – $\mathcal{T}_{\mathrm{IE}}$

$\mathcal{T}_{\mathrm{IE}}$

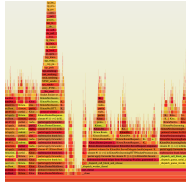
Sunburst View



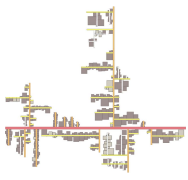
BeamTree



Bundle View



Flame Graphs

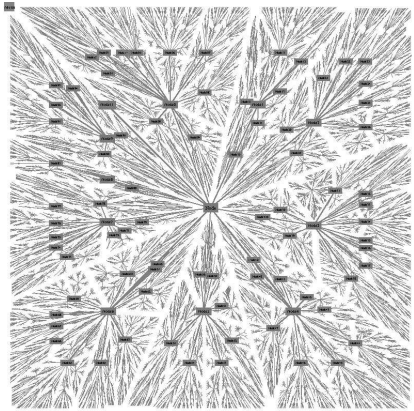


Software Cities

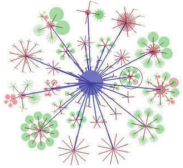
Examples of treemaps from the category *Implicit Edge-Representation Treemap*. Image courtesy from left to right and top to bottom: [Stasko et al., 2000], [van Ham and van Wijk, 2003], [Holten, 2006], [Gregg, 2016], and [Steinbrückner and Lewerentz, 2013].

Mapped Tree – $\mathcal{T}_{\mathrm{MT}}$

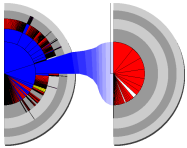
$\mathcal{T}_{\mathrm{MT}}$



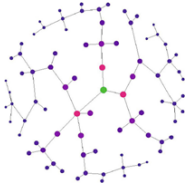
Space-optimized Tree



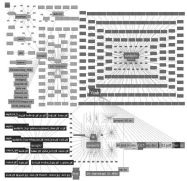
RINGS



Information Slices



Tree Drawings

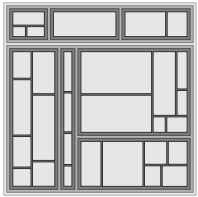


ENCCON

Examples of treemaps from the category *Mapped Tree*. Image courtesy from left to right and top to bottom: [Nguyen and Huang, 2002], [Tee Teoh and Kwan-Liu, 2002], [Andrews and Heidegger, 1998], [Toosi and Nikolov, 2014], and [Nguyen and Huang, 2005].

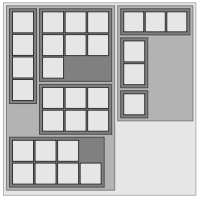
Proposed Categories of Treemap Taxonomy

$\mathcal{T}_{\mathrm{S}}$



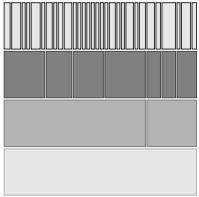
Space-filling Treemap

$\mathcal{T}_{\mathrm{C}}$



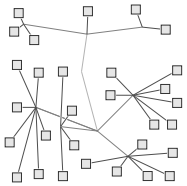
Containment Treemap

$\mathcal{T}_{\mathrm{IE}}$



Implicit Edge Representation Tree

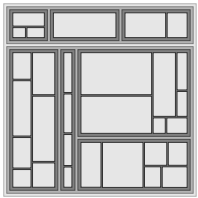
$\mathcal{T}_{\mathrm{MT}}$



Mapped Tree

Proposed Categories of Treemap Taxonomy

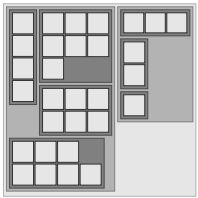
$\mathcal{T}_{\mathrm{S}}$



Space-filling Treemap

$\subset$

$\mathcal{T}_{\mathrm{C}}$



Containment Treemap

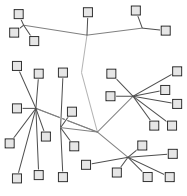
$\subset$

$\mathcal{T}_{\mathrm{IE}}$

Implicit Edge Representation Tree

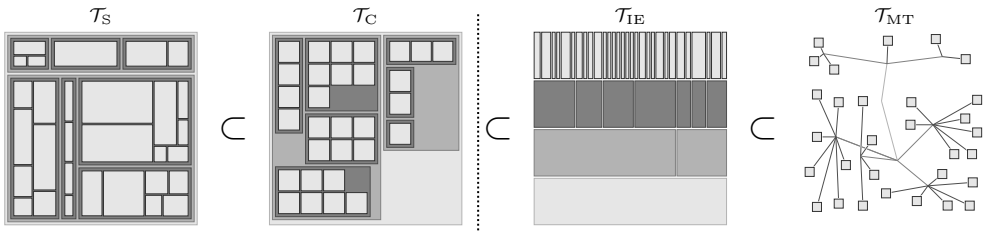
$\subset$

$\mathcal{T}_{\mathrm{MT}}$



Mapped Tree

Proposed Categories of Treemap Taxonomy



The four proposed treemap categories, namely Space-filling Treemap, Containment Treemap, Implicit Edge Representation Tree, and Mapped Tree. Each category includes their predecessors, building a more general category. The dotted line highlights the observed current border, whether a visualization technique is called a treemap.

DISCUSSION
discovery
gateway
local
search
type

Insights

- Aim at the most specific category
- All visualization techniques covered by this taxonomy use an underlying treemap layout
- Only a systematic top-down or bottom-up approach adhering to the containment metaphor results in a treemap

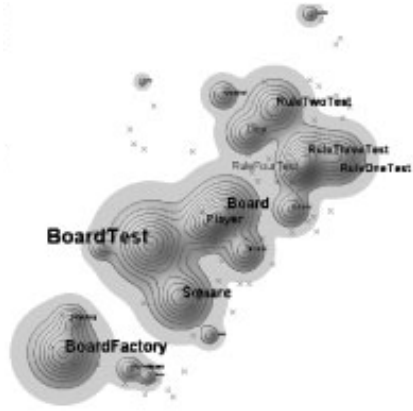
Limitations

- Taxonomy does not cover mixed-use of treemaps and basic forms of subdivision
- Taxonomy does not cover all tree visualization techniques
- Conceptual border between mapped trees and general tree and hierarchy visualization techniques to be determined

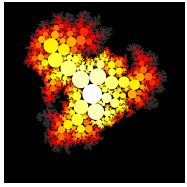
Adjacent placement of multiple treemap visualizations.

Comparison to General Hierarchy Visualization Techniques

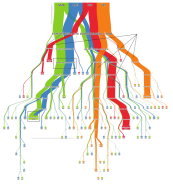
$\mathcal{H}$



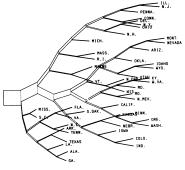
Thematic Software Maps



Bubble Hierarchies



BaobabView



Tree of Percentage

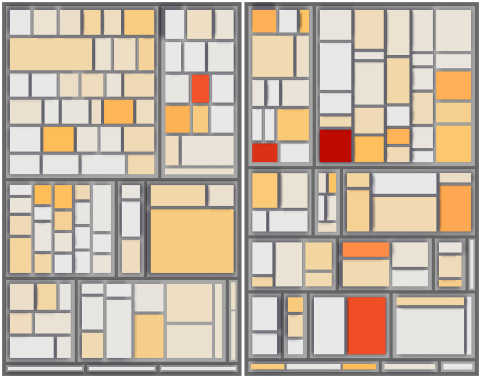
Pythagoras Tree

Examples of visualization techniques that does not relate to any proposed class and are therefore associated with $\mathcal{H}$. Image courtesy from left to right and top to bottom: [Kuhn et al., 2010], [Hlawatsch et al., 2014], [Andrews and Heidegger, 1998], [van den Elzen and van Wijk, 2011], [Kleiner and Hartigan, 1981], and [Beck et al., 2014].

CONCLUSIONS

Conclusions

- We propose a taxonomy of treemap visualization techniques using four distinctive classes:
  - $\mathcal{T}_{\mathrm{S}}$ Space-filling Treemap
  - $\mathcal{T}_{\mathrm{C}}$ Containment Treemap
  - $\mathcal{T}_{\mathrm{IE}}$ Implicit-Edge Representation Tree
  - $\mathcal{T}_{\mathrm{MT}}$ Mapped Tree
- List of properties for each class and the term treemap in general
- Exemplary assignment of existing treemap and general tree visualization techniques



Future Work

- Refinement of proposed taxonomy
  - Evaluation of more tree visualization techniques
  - Application in tree visualization libraries and software systems
- Broadening of the concept of treemaps for tree and hierarchy visualization

A Taxonomy of Treemap Visualization Techniques

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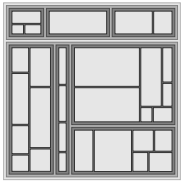
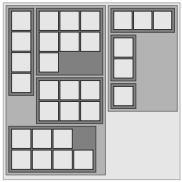
Computer Graphics Systems Group at Hasso Plattner Institute, Digital Engineering Faculty, University of Potsdam
(https://hpi3d.de)

$\mathcal{T}_{\mathrm{S}}$



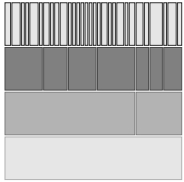
Space-filling Treemap

$\mathcal{T}_{\mathrm{C}}$



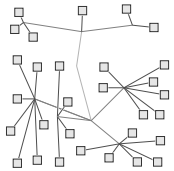
Containment Treemap

$\mathcal{T}_{\mathrm{IE}}$

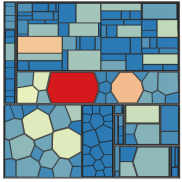


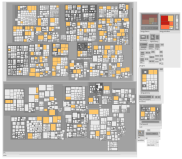
Implicit Edge Representation Tree

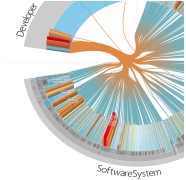
$\mathcal{T}_{\mathrm{MT}}$

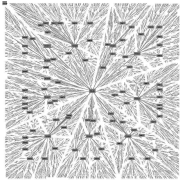


Mapped Tree









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